Zero Incident – Safety Training Simulator

Prototype Operation Guide (README)

Module: CET257 – Enterprise Project
Assessment: Assignment 1 (A5, A6 & A7)
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1. Purpose of the Prototype

This prototype demonstrates the **core interaction, usability, and engagement concepts** of the Zero Incident Safety Training Simulator — a 360° warehouse safety-training system.

It is a **proof of concept**, not a full production deployment.

Two entry points are provided: an interactive offline prototype (*demo.html*) and the fully working system (Node.js).

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2. System Requirements

- Desktop or laptop computer
- Modern web browser (Chrome, Edge, Firefox)
- No internet connection required for the prototype
- For the full system: Node.js 20 or later (no external database required)

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3. How to Launch

Option A — Interactive prototype (no setup)

1. Open the project folder
2. Double-click *demo.html*
3. The prototype opens automatically in a web browser

Option B — Full system

1. Open PowerShell (or a terminal) in the project folder
2. Run *npm install*
3. Copy *.env.example* to *.env*
4. Run *npm start*
5. Open http://localhost:3000

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4. Training Flow Overview

The prototype walks through a complete training journey:

- Login (employee and administrator roles)
- Walkable 3D warehouse training hub
- Module briefing from the 3D trainer
- 360° practical activity (checkpoints / hazard identification)

- Timed knowledge quiz
- Scoring and pass/fail decision
- Retake route and completion certificate

Navigation is controlled with on-screen controls (WASD + mouse or the on-screen d-pad) and buttons.

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5. Operator Instructions

- Launch *demo.html* (or the full system)
- Sign in as an employee (demo account *ramesh.s* / *Welcome!2026* — auto-filled in the prototype)
- Enter the 3D Training Hub and open a module door (Hazard Perception, Manual Handling, Working at Height)
- Complete the practical checkpoints in the 360° scene
- Answer the timed quiz questions
- View your score; retake the module if below 70%, or view the completion certificate
- Administrator (full system): manage employees, monitor progress, and review results

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6. Scoring & Feedback

- Practical checkpoints and quiz questions earn marks
- An overall score of 70/100 is required to pass a module
- Instant feedback with an explanation is shown for every answer
- Retakes are allowed; the best assessed score is kept
- A completion certificate is issued for passed modules

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7. Technical Notes

- *demo.html* is built with HTML, CSS, and JavaScript — fully browser-based
- No server, database, or login required for the prototype
- Full system: Node.js and Express, MongoDB (bundled in-memory), Three.js 3D/360° engine
- Role-based access control (RBAC) with session and CSRF protection
- Certificates are generated as PDF
- Designed to support future VR expansion

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8. Limitations

- The 360° environment is a simulation, not full VR
- Limited number of scenarios
- Some features remain in progress (badges, advanced analytics, final module interiors)
- The prototype does not include persistent data storage

These limitations are expected at prototype stage.

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9. Academic Context

Developed for the CET257 Enterprise Project to demonstrate:

- User-centred design
- Interactive 3D training concepts
- Client-ready prototype presentation

End of ReadMe

Thank You